

Extrema and the Mean Value Theorem



Critical points and extrema

- 1 Find the critical points of $f(x) = 2x^3 - 3x^2 - 12x + 5$.
 - 2 Find the absolute maximum and minimum of $f(x) = x^3 - 3x$ on $[-2, 3]$, checking critical points and endpoints.
-

The Extreme Value Theorem

- 3 State the Extreme Value Theorem and explain why it guarantees $f(x) = x^2 - 4x$ has an absolute max and min on $[0, 5]$.
-

Rolle's Theorem

- 4 Verify that Rolle's Theorem applies to $f(x) = x^2 - 4x + 3$ on $[1, 3]$, and find the value(s) of c guaranteed by the theorem.
 - 5 Explain why Rolle's Theorem does NOT apply to $f(x) = |x - 2|$ on $[0, 4]$, even though $f(0) = f(4) = 2$.
-

The Mean Value Theorem

- 6 For $f(x) = x^3$ on $[0, 2]$, find the value(s) of c guaranteed by the Mean Value Theorem.
 - 7 A car travels 180 km in 2 hours. Using the Mean Value Theorem, explain why the car must have been traveling at exactly 90 km/h at some instant.
-

Using the MVT to bound function values

- 8 Given $f(2) = 5$ and $-3 \leq f'(x) \leq 4$ for all x , use the MVT to find the range of possible values for $f(6)$.
-

Consequences of the MVT: monotonicity

- 9 Explain, using the MVT, why $f'(x) > 0$ on an interval implies f is strictly increasing on that interval.

- 10 Explain why two functions f and g with $f'(x) = g'(x)$ for all x must differ by a constant, using the MVT.